

**24** A rectangular coil of wire lies in a uniform magnetic field of 0.30 T. The field is perpendicular to the plane of the coil. The coil is stretched from dimensions 0.20 m × 0.50 m to 0.20 m × 0.80 m in 2.0 seconds.

What is the average emf induced in the coil during this time?

**A**

0.009 V

**B**

0.018 V

**C**

0.027 V

**D**

0.036 V

**Answer: A**

Initial area =  $0.20 \times 0.50 = 0.10 \text{ m}^2$

Final area =  $0.20 \times 0.80 = 0.16 \text{ m}^2$

Change in flux,  $\Delta\Phi = B \times \Delta A = 0.30 \times (0.16 - 0.10) = 0.018 \text{ Wb}$

Average emf =  $\Delta\Phi / \Delta t = 0.018 / 2.0 = 0.009 \text{ V}$

**[Turn over**

**25** The variation of an alternating voltage  $V$  with time  $t$  is shown in the graph below